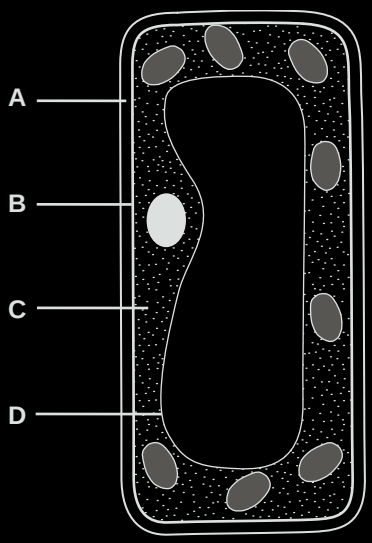
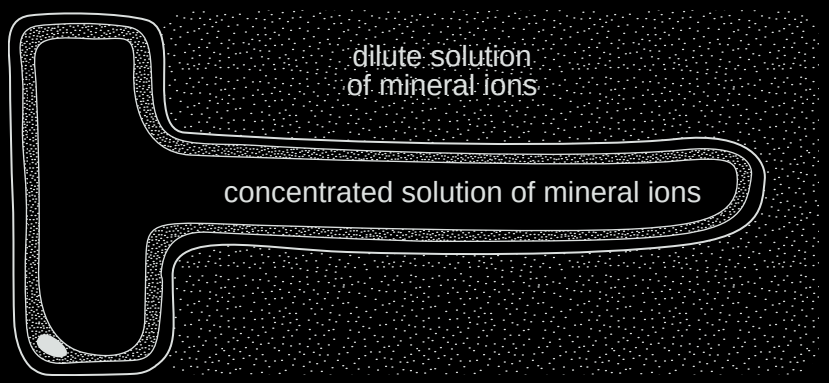


2 The diagram shows a root hair, surrounded by a dilute solution of mineral ions.



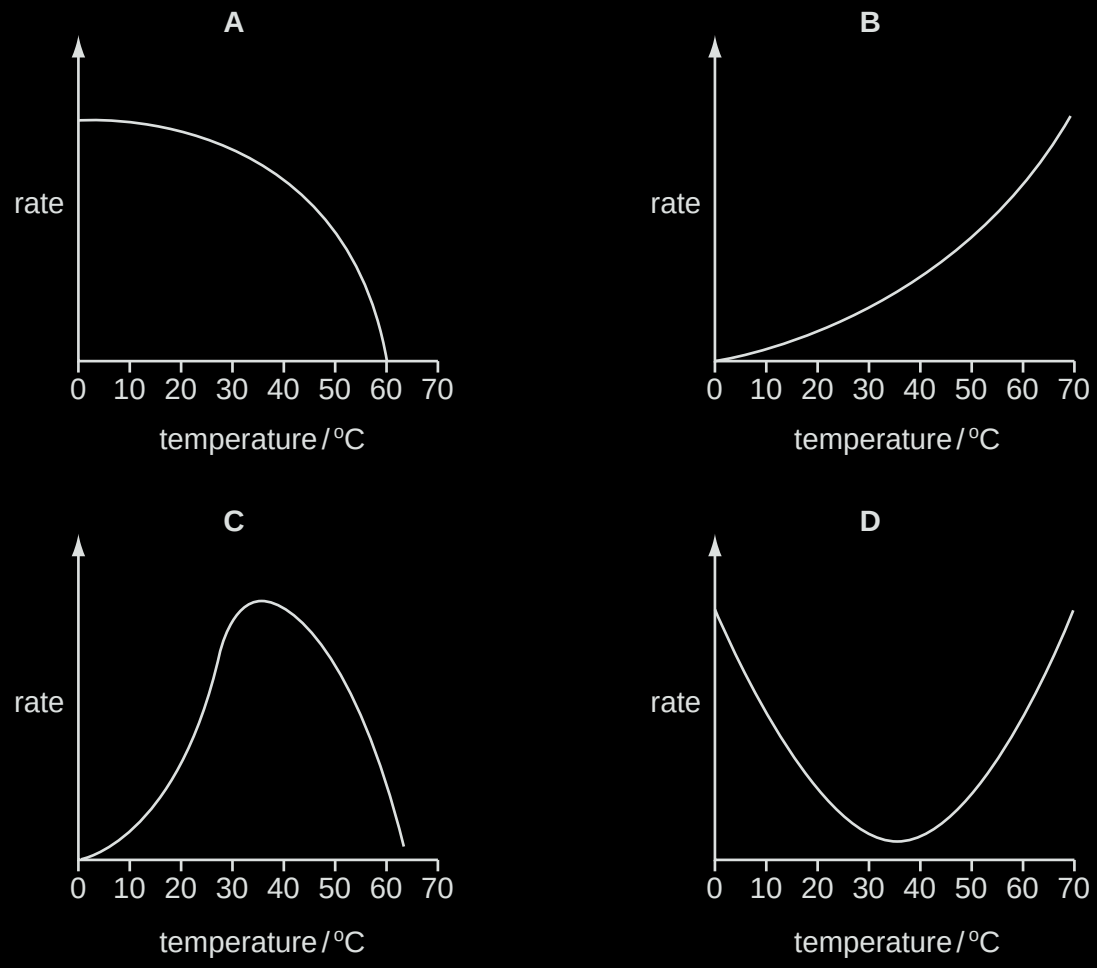
Which statement is correct?

A Water molecules move into the root hair because their concentration is lower inside.

B Water molecules move into the root hair because their concentration is lower outside.

C Water molecules move out of the root hair because their concentration is lower inside.

D Water molecules move out of the root hair because their concentration is lower outside.



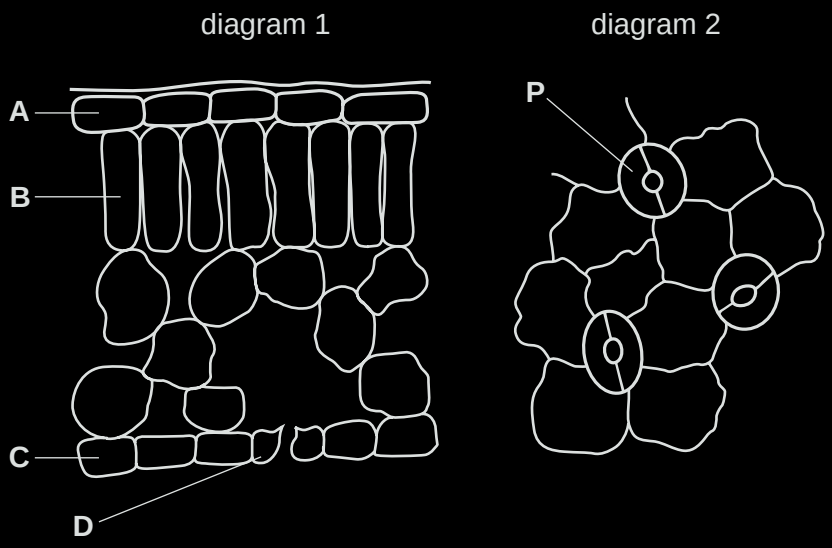
4 Which process can involve active transport?

A carbon dioxide intake through stomata

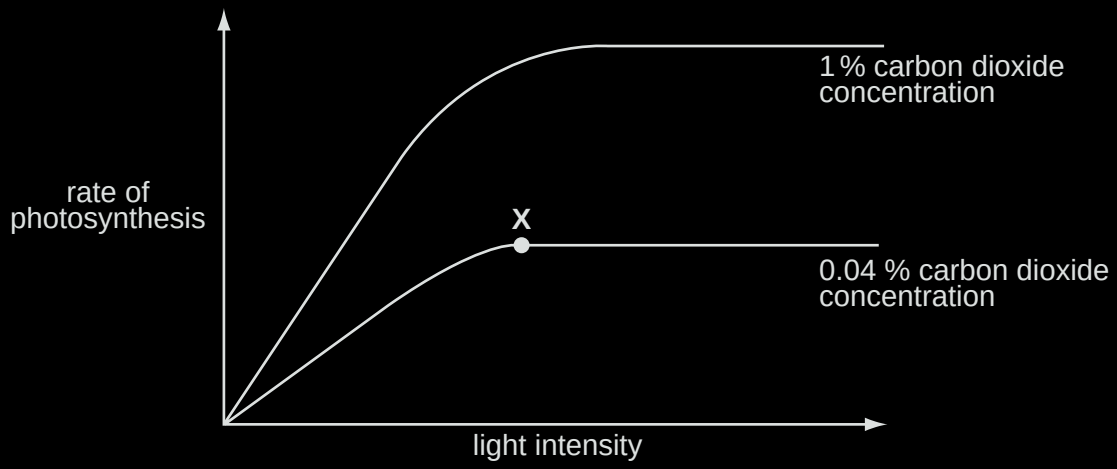
B mineral ion intake through root hairs

C mineral ion transport through xylem vessels

D water leaving mesophyll cells



6 The graph shows how the rate of photosynthesis of a plant varies with light intensity at two different carbon dioxide concentrations. The temperature is kept constant at 20 °C.



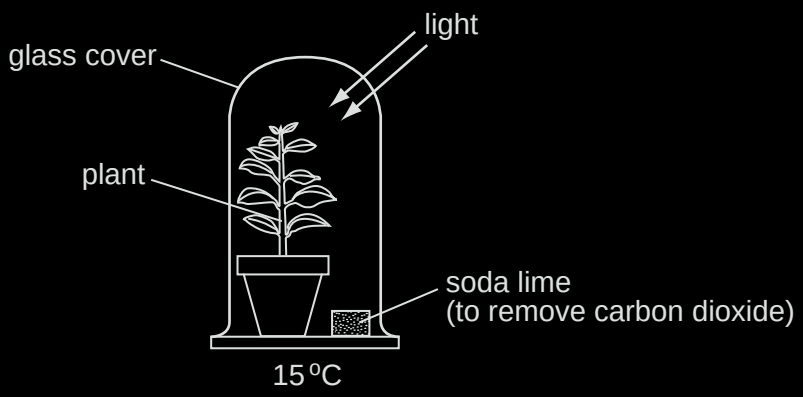
Which factor is limiting the rate of photosynthesis at point **X**?

A availability of chlorophyll

B availability of water

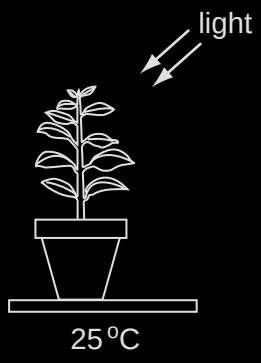
C concentration of carbon dioxide

D intensity of light

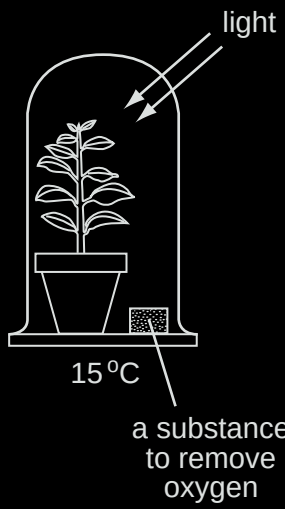


What is the most suitable control for this experiment?

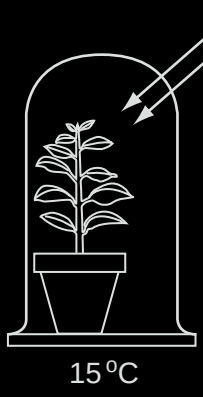
A



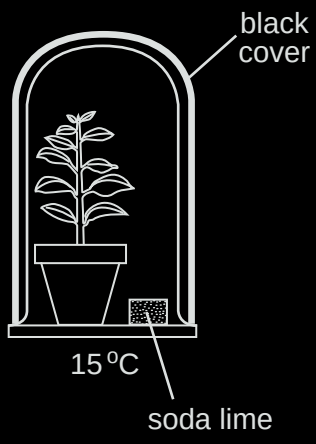
B

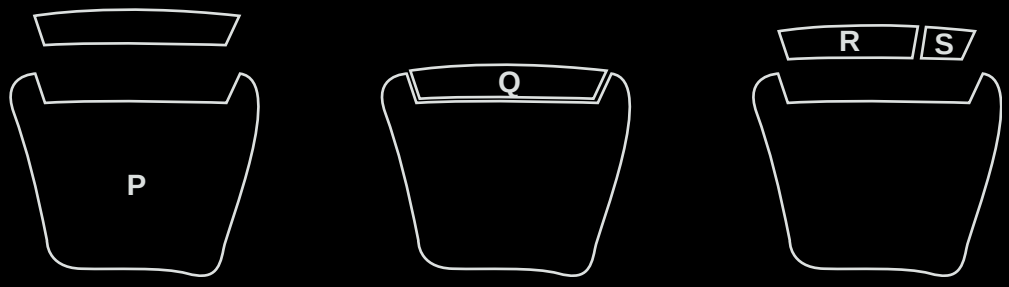


C



D





Which are the correct labels?

	amylase	maltose	starch
A	**P**	**S**	**Q**
B	**Q**	**R**	**S**
C	**R**	**Q**	**P**
D	**S**	**P**	**R**

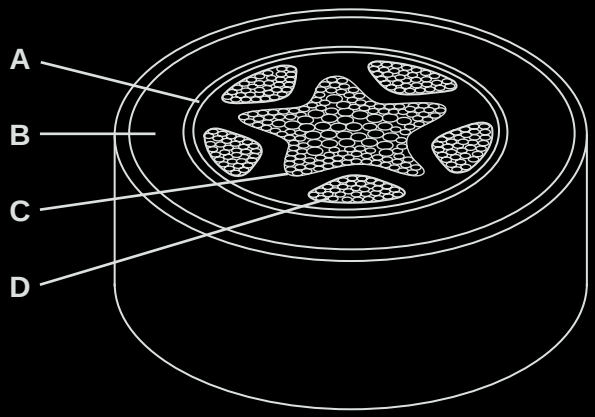
11 Which conditions cause the fastest rate of transpiration?

A dry and cold

B dry and warm

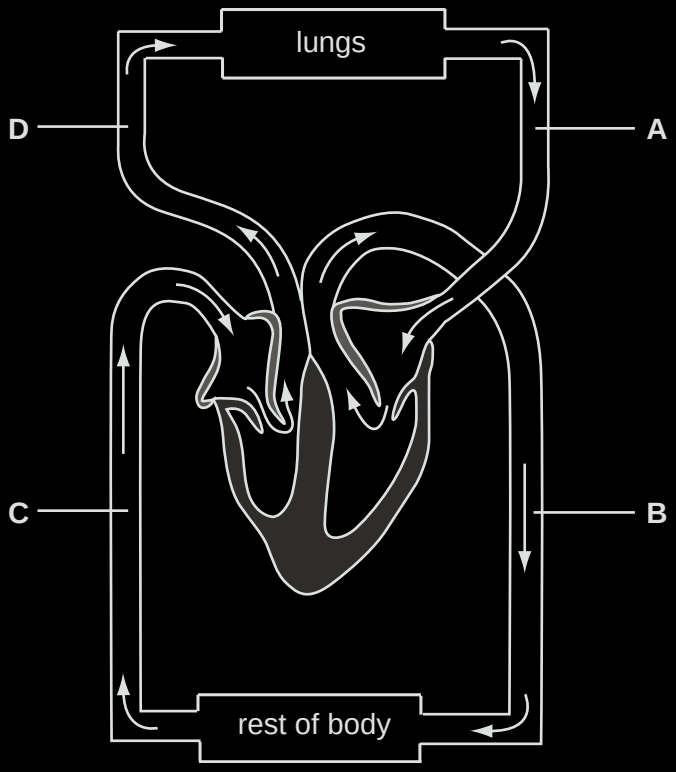
C wet and cold

D wet and warm



13 The diagram shows part of the human circulatory system.

In which vessel is the blood pressure highest?



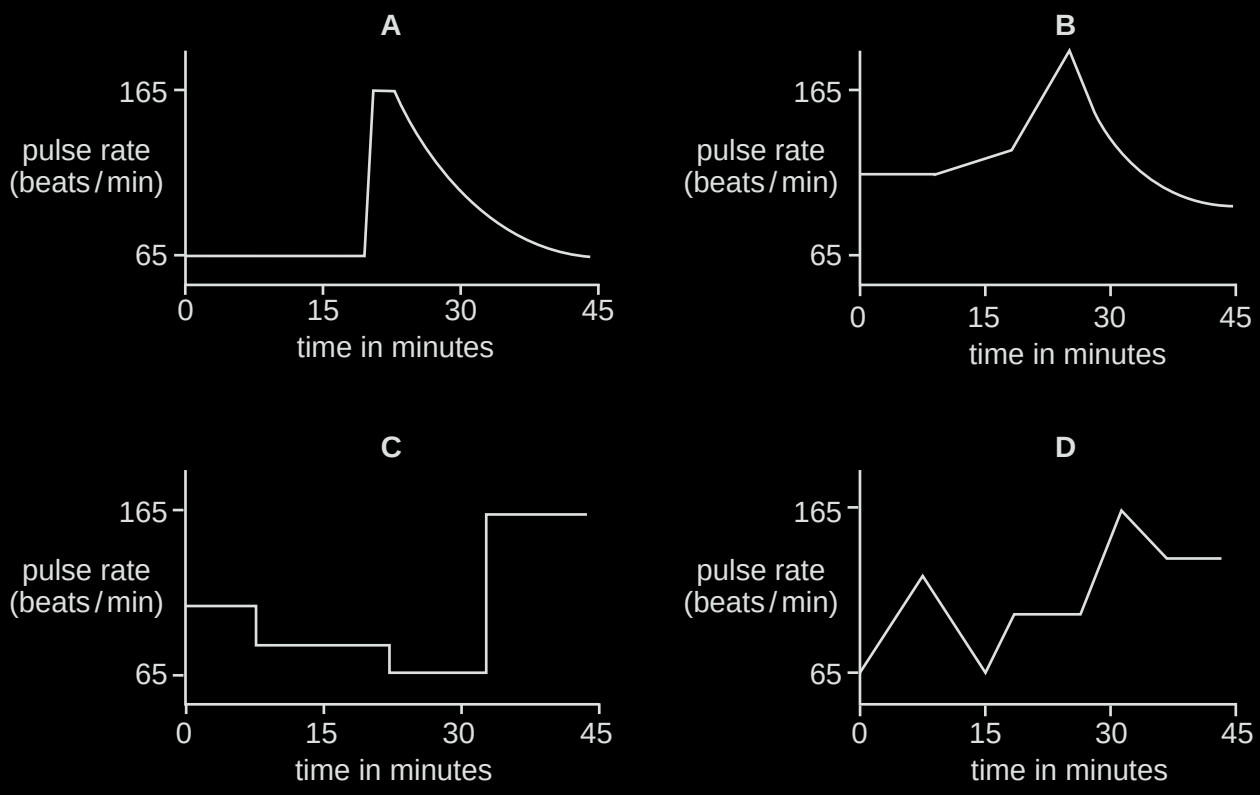
14 Which blood vessel carries blood with the **lowest** concentration of urea?

A hepatic portal vein

B pulmonary vein

C renal vein

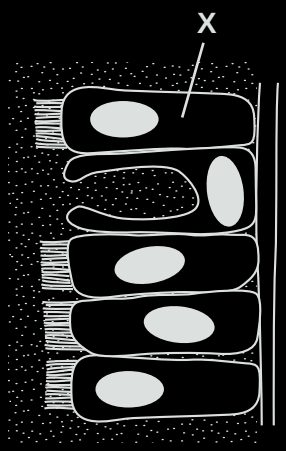
D vena cava



16 When the skin is cut the blood clots.

In which order would the components of the blood become involved?

	first →			last
A	fibrin	platelets	red blood cells	fibrinogen
B	fibrinogen	red blood cells	platelets	fibrin
C	platelets	fibrin	fibrinogen	red blood cells
D	platelets	fibrinogen	fibrin	red blood cells



What is the function of cell **X**?

A gaseous exchange

B mucus removal

C phagocytosis

D secretion of mucus



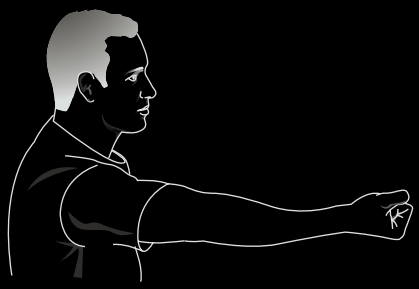
The ball and socket joint rotates and the hinge joint extends.

Which diagram shows the arm after these movements?

A



B

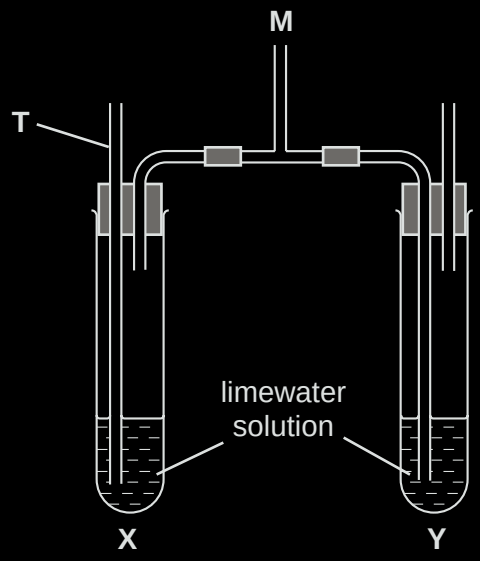


C



D





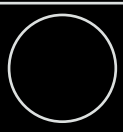
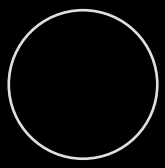
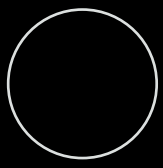
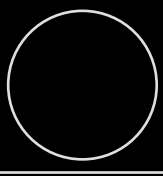
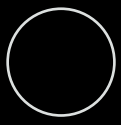
What would occur when a person breaths out through tube **M**?

A The solutions in **X** and **Y** both turn cloudy.

B The solution in **X** remains clear, but that in **Y** turns cloudy.

C The solution in **X** turns cloudy, but that in **Y** remains clear.

D The solution in **X** is forced out through the tube **T**.

	diameter of blood vessels	
	after a decrease in body temperature	after an increase in body temperature
A		
B		
C		
D		

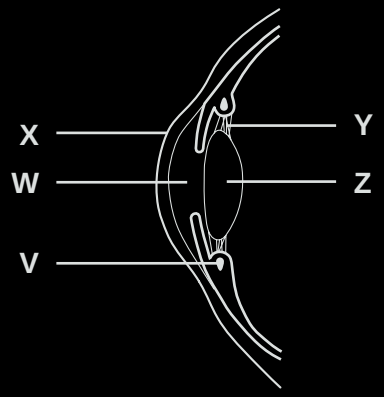
22 What passes through the membranes of a kidney machine?

A protein and red blood cells

B urea and red blood cells

C water and protein

D water and urea



Which parts will change to focus on a distant object?

A	**W**	**X**	**Y**
B	**V**	**X**	**Z**
C	**W**	**Y**	**Z**
D	**V**	**Y**	**Z**

24 The table shows results from a study into the effects of smoking while pregnant.

number of cigarettes smoked per day by mother while pregnant	average birth weight of baby / kg	average height of child at 15 years / cm
0	3.7	166.1
1–9	3.5	165.0
10 and over	3.1	162.8

Which of these effects of smoking while pregnant is supported by the information in the table?

A increased growth rate

B increased length of pregnancy

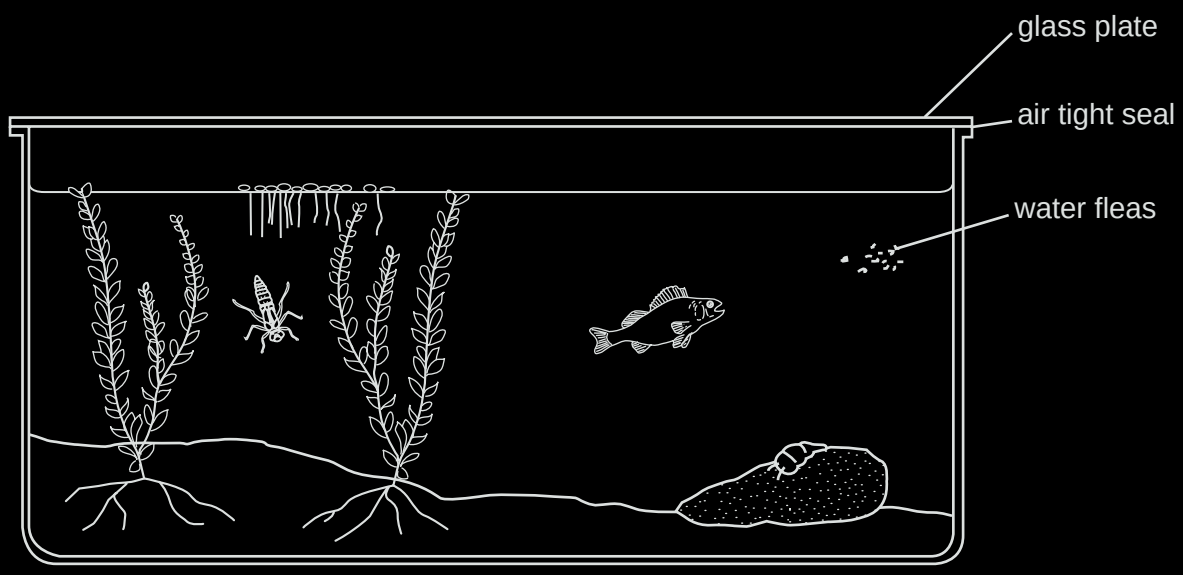
C reduced birth weight

D reduced chance of respiratory disease



Which regions control balance, heart rate and temperature?

	balance	heart rate	temperature
A	**W**	**Z**	**X**
B	**X**	**Y**	**Z**
C	**Y**	**X**	**W**
D	**Z**	**W**	**Y**



What must be supplied to keep the organisms alive for the longest possible time?

A carbon dioxide

B light energy

C nitrates

D oxygen

28 What is a feature of bacteria?

A They are always parasitic.

B They have a nucleus.

C They are made of hyphae.

D They reproduce by binary fission.

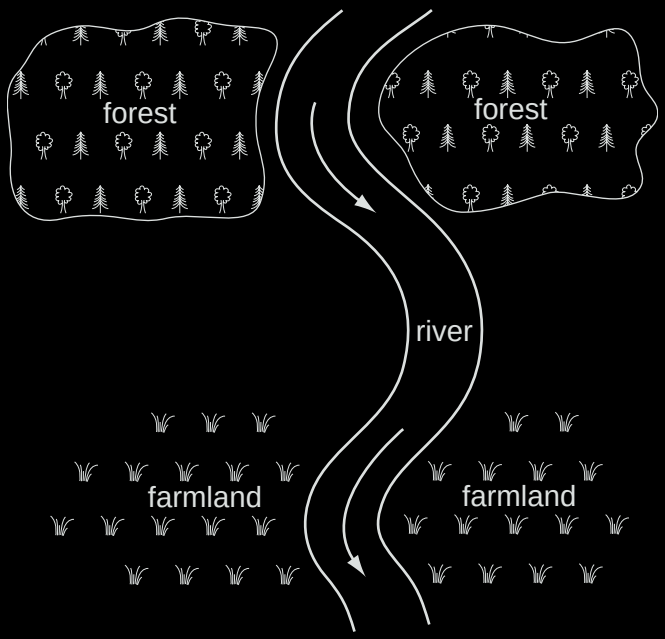
29 Which method of control would **not** be effective against the spread of the malarial parasite?

A drainage of swamps and marshes

B safe disposal of sewage solids

C sleeping under a mosquito net

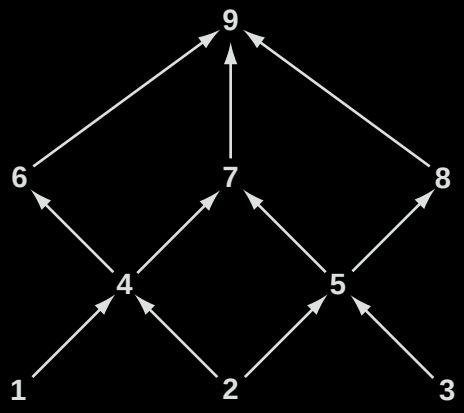
D spraying walls of houses with insecticide



The forest is cut down.

What will be the most likely effect and how it is connected to the cutting down of the forest?

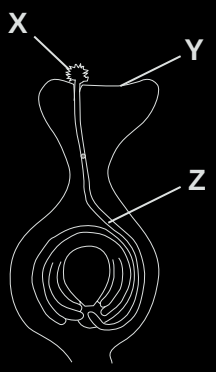
	likely effect	how it is connected to the cutting down of the forest
A	flooding of the farmland	water running off cleared area
B	gradual change to desert conditions	global warming
C	increase in number of water plants	more light falling on river
D	water logging of cleared area	trees no longer transpiring



Which of the organisms is a producer and which is a carnivore?

	producer	carnivore
A	**1**	**4**
B	**2**	**6**
C	**5**	**8**
D	**7**	**9**

33 The diagram shows the carpel of a flower soon after pollination.



What are the labelled structures?

	X	**Y**	**Z**
A	ovule	stigma	filament
B	ovule	stamen	pollen tube
C	pollen grain	stigma	pollen tube
D	pollen grain	stamen	filament